
AiO-MM: Agentic Decomposition and Distillation for Multifaceted Multimodal Content Understanding

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Abstract

1 Understanding complex multimodal content on social media is inherently multi-
2 faceted, requiring simultaneous analysis across figurative, emotional, and social-
3 harm dimensions. Yet existing work remains fragmented, with isolated subtask
4 formulations and disparate datasets. We introduce **AiO-MM**, a unified three-stage
5 framework that addresses this challenge via (i) zero-shot multifaceted analysis
6 through a collaborative multi-agent system (**AiO-MMAgents**) that decomposes
7 content understanding into interpretable agent roles grounded in figurative lan-
8 guage, (ii) **Human-AI Annotation** to curate a high-quality unified multi-task
9 dataset (**Multifacet-MM**), and (iii) **Agentic Distillation** that transfers structured
10 multi-agent reasoning into an efficient student model. We instantiate and eval-
11 uate our framework on internet memes—a challenging domain where figura-
12 tive language (metaphor, sarcasm, irony) interacts with visual elements to pro-
13 duce nuanced social meaning. Experiments demonstrate strong effectiveness:
14 AiO-MMAgents achieves average scores of 71.16% on HatReD and 74.59%
15 on MET-Meme, outperforming few-shot chain-of-thought prompting by up to
16 13.2 points, while the distilled AiO-MM retains competitive performance at
17 substantially reduced computational cost. Our code and data are available at
18 <https://anonymous.4open.science/r/DeDi>.

19 1 Introduction

20 Multimodal content on social media is inherently multifaceted, demanding simultaneous analysis
21 across figurative, emotional, and social-harm dimensions. Yet existing approaches to multimodal
22 content understanding address these dimensions in isolation, with disparate task definitions and
23 incompatible annotation schemes, limiting both cross-task generalization and holistic analysis. Inter-
24 net memes Pramanick et al. [2021], Lin et al. [2024b] offer a particularly challenging instantiation
25 of this problem: they combine images and text to convey humor, commentary, and sometimes
26 harmful messages through figurative expressions such as metaphor, sarcasm, and irony. Prior work
27 in Computational Meme Understanding (CMU) Nguyen and Ng [2024] has addressed individual
28 aspects—hateful content detection, sentiment classification, or figurative language recognition—but
29 a unified, multifaceted perspective remains elusive.

30 Prior efforts toward multi-task analysis have proposed joint CMU frameworks Chauhan et al. [2020],
31 Zheng et al. [2025], Xu et al. [2022], Maity et al. [2022], Sharma et al. [2023], Zhong and Baghel
32 [2024], but each covers only a subset of dimensions (Table 1). For example, MET-Meme Xu
33 et al. [2022] addresses sentiment, intensity, and intention with a focus on metaphor, while HatReD
34 Kiela et al. [2020] targets social category, target, and attack type—with no single work unifying
35 all dimensions. Critically, none of these approaches leverage the structured reasoning capabilities
36 of modern multi-agent systems, nor address how to transfer such reasoning efficiently to smaller,
37 deployable models.

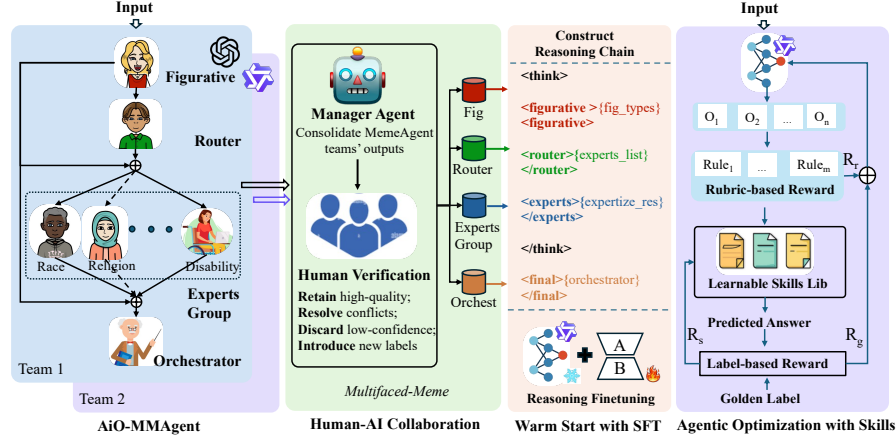


Figure 1: Overview of our proposed 3-stage pipeline **AiO-MM**. 1) Zero-shot Multifaceted CMU uses a multi-agent framework **AiO-MM Agents**; 2) Human-AI Collaboration for Annotation to curate a high-quality dataset **Multifacet-MM** (Section 3.2); 3) Reasoning Finetuning as warm start to transfers the reasoning patterns of agentic collaboration into a student model; 4) Agentic Optimization with learnable skills using GRPO (Section 3.3).

Table 1: Dataset statistics and comparison of prior multi-task CMU studies. We aim to provide a unified work for multifaceted multimodal content understanding using memes as a testbed. * indicates the source datasets used in this study.

Dataset and/or Publication	#Task-Sample Pairs	Objective	#Sample	Figurative	Sentiment	Intention	Intensity	Social Category	Social Target	Attack Type	Explanations
WOAH-5* Mathias et al. [2021]	20,000	Hate	10,000					✓		✓	
MET-Meme* Xu et al. [2022]	40,180	General	10,045	Metaphor	✓	✓	✓				
FigMeme Liu et al. [2022]	5,141	General	5,141	Multi-class							
HatReD* Hee et al. [2023]	12,912	Hate	3,228					✓	✓	✓	✓
MemeIntent Park et al. [2024]	1,900	General	950			✓					✓
Multifacet-MM (Ours)	26,400	Mix	3,300	Multi-class	✓	✓	✓	✓	✓	✓	✓

In this paper, we address this gap with **AiO-MM**, a unified three-stage framework built around *agentic decomposition and distillation* for multifaceted multimodal content understanding. The key insight is that complex multimodal analysis can be decomposed into interpretable agent roles—figurative understanding, adaptive routing, expert analysis, and orchestration—whose structured reasoning can subsequently be distilled into an efficient student model. We instantiate and evaluate AiO-MM on internet memes as a demanding testbed, identifying eight representative CMU tasks (Table 2) that collectively span figurative, emotional, and social-harm dimensions.

We organize these tasks into three groups and address them with **AiO-MM Agents**, a collaborative multi-agent system comprising four core components: a Figurative Agent, a Router, an Experts Group, and an Orchestrator Agent. To bridge the absence of a unified multi-task dataset, we introduce a Manager Agent OpenAI [2024], Subramaniam et al. [2025] that synthesizes outputs from two AiO-MM Agents teams with background knowledge from existing datasets, supporting scalable Human-AI collaborative annotation. Finally, we transfer the structured reasoning of the full agentic pipeline into a compact student model via reasoning-chain supervision and reinforcement learning with hybrid reward signals Huang et al. [2023], Hu et al. [2021]. Our main contributions are as follows:

- We propose **AiO-MM**, a unified three-stage framework for multifaceted multimodal content understanding via agentic decomposition, Human-AI annotation, and agentic distillation, instantiated on internet memes as a challenging testbed.
- We design **AiO-MM Agents**, a zero-shot collaborative multi-agent system that grounds multifaceted analysis in figurative language understanding.
- We curate **Multifacet-MM**, a Human-AI annotated dataset with unified labels across eight tasks spanning figurative, emotional, and social-harm dimensions.
- We propose an **Agentic Distillation** strategy that transfers structured multi-agent reasoning into an efficient Qwen3-VL-8B student model via reasoning-chain supervision and GRPO-based optimization.

Table 2: Definition of Multifaceted CMU, unifying eight tasks from prior work (Table 1). C denotes classification tasks and G denotes generation tasks.

Task	Name	Type	Option List
T1	Figurative	C	metaphor, irony, hyperbole, onomatopoeia, personification, allusion, idiom, pun
T2	Social Category	C	gender, race, age, religion, disability, nationality, none
T3	Social Target	G	–
T4	Sentiment	C	happiness, love, anger, sorrow, fear, hate, surprise, neutral
T5	Intention	C	interactive, expressive, purely entertaining, offensive, other
T6	Intensity	C	integer scores ranging from 1 (very mild) to 4 (intense)
T7	Attack Type	C	dehumanizing, inferiority, inciting violence, mocking, contempt, slurs, exclusion, none
T8	Explanation	G	–

2 Related work

Multi-task CMU Nguyen and Ng [2024] argue that CMU encompasses a wide range of tasks, including figurative language interpretation and fine-grained emotion recognition. A large body of work focuses on harmful meme classification Gupta et al. [2021], Cao et al. [2022], Bianchi et al. [2022], Zhang et al. [2023a], Lin et al. [2023], Wen et al. [2023], Ocampo et al. [2023], Lin et al. [2024b], while another line of research explores multi-task CMU. For example, Sharma et al. [2020] introduce a multi-task challenge covering sentiment classification, emotion type recognition, and intensity prediction. Although several datasets support multi-task CMU Mathias et al. [2021], Xu et al. [2022], Shah et al. [2024], differences in task definitions and coverage complicate cross-task integration and limit comprehensive multi-task analysis

Figurative language (e.g., metaphor and irony) plays a central role in making memes engaging and widely shared Paul [1970], Yosef et al. [2023], Sharma et al. [2023], Zhong and Baghel [2024], Liu et al. [2022], Hee and Lee [2025]. Prior work shows that metaphor comprehension can enhance emotion recognition Zhang et al. [2024]. However, visual metaphors remain challenging for multi-modal large language models (MLLMs) Hwang and Shwartz [2023], though reasoning strategies such as Chain-of-Thought (CoT) prompting Wei et al. [2023] can improve metaphor detection Xu et al. [2024]. Other figurative phenomena, such as sarcasm and irony, have also been studied Lin et al. [2024a], Chauhan et al. [2020], and several datasets explicitly target figurative language in memes Yosef et al. [2023], Liu et al. [2022]. Despite these efforts, the role of multimodal figurative language understanding in enabling comprehensive CMU remains underexplored.

There have been some efforts toward developing all-in-one frameworks for multi-task CMU. Early work includes joint modeling of multiple meme attributes, such as humor, sarcasm, offensiveness, motivation, and sentiment Chauhan et al. [2020]. Subsequent approaches are dataset-specific and based on relatively small foundation models, including BERT-based text encoders with VGG or ResNet vision backbones Maity et al. [2022], Shah et al. [2024]. In this work, we build on these efforts and propose an all-in-one multi-agent framework for multi-task CMU.

Knowledge Distillation Knowledge distillation Hinton et al. [2015] is a standard approach for transferring knowledge from large teacher models to compact student models. Recent studies extend this paradigm by using LLMs and multi-agent systems as teachers that provide rich reasoning traces for training efficient models Ravi et al. [2024], Zhao et al. [2024], Li et al. [2024], Chen et al. [2024]. Ng [2025] formalize this setting as *agentic distillation*. To our knowledge, this paradigm has not been explored for multifaceted computational meme understanding, where both reasoning quality and deployability are critical.

3 Method

We describe **AiO-MM** in this section, a three-stage pipeline grounded in multi-agent collaboration and figurative language understanding for unified multifaceted CMU. We hypothesize that multimodal figurative reasoning contributes to downstream sub-tasks, a claim supported by the ablation results shown in Figure 5.

3.1 A Multi-Agent Framework for CMU

Let $D = \{x_1, x_2, \dots, x_n\}$ be a dataset of n memes, where each meme $x \in D$ has an image x_v and caption x_t . The goal of **AiO-MMAGents** is to perform zero-shot, multi-task CMU on this input. While Huang et al. [2024] introduced an agent-based framework for harmful meme classification, it targets a specific CMU setting and does not consider figurative language or broader multifaceted modeling. We extend this line of work by proposing a novel agentic framework for Multifaceted CMU that supports both harmful and general memes. We decompose the problem into 4 task layers and design the AiO-MMAGents architecture according to the task-level Mixture-of-Experts (MoE) principle Jiang et al. [2024], DeepSeek-AI et al. [2024]. The roles and design of the four task-specific agents are described below.

Figurative Agent We design a Figurative Agent, denoted as A_f , to serve as a shared expert for task T1: Figurative Language Understanding. We hypothesize that figurative expressions play a key role in uncovering the deep implicit semantics embedded in memes and can significantly contribute to downstream tasks T2-T8. Let p_f represent the prompt for the Figurative Agent (see Appendix A for the prompt details), and let m be the backbone MLLM used by agent. The output generated by the agent is denoted as: $y_f = A_f(x_v, x_t, p_f, m)$ (1).

Router We design a Router A_r with prompt p_r for T2, Social Category Classification Hee et al. [2023], based on the analysis from Figurative Agent y_f . Let $\mathcal{C} = \{\text{gender, race, age, religion, disability, nationality}\}$ be the set of social categories. The Router determines if a social category $s \in \mathcal{C}$ is potentially involved in or affected by the meme. The decision will help to narrow down the social dimensions relevant to the meme’s content and activate expert agents for more focused and efficient downstream evaluation. The Router is training-free and operates via prompt-driven routing, deriving its decision-making capability directly from the shared backbone MLLM rather than from a separately optimized classifier: $y_r = A_r(y_f, x_v, x_t, p_r, m)$, where $y_r \subseteq \mathcal{C}$ (2).

Experts Group Building on the framework proposed by Zhang et al. [2023b], we design an Expert Group A_g , composed of 6 specialized expert agents. The group is responsible for conducting fine-grained, multi-task analysis of memes (Task T3–T8), guided by the set of activated social categories y_r identified by the Router agent. We define the set of category-specific prompts as $\mathcal{P}_\mathcal{C} = \{p_c \mid c \in \mathcal{C}\}$, where each p_c corresponds to the prompt used by the expert agent assigned to analyze the meme from the perspective of social category c . For each activated social category $c \in y_r$, we retrieve the corresponding prompt p_c and activate the relevant expert A_g to perform downstream tasks T3 ~T8. This analysis leverages the shared figurative understanding y_f and the original meme inputs (x_v, x_t) . The output of the expert agent for category c is denoted as: $y_c = A_g(y_f, x_v, x_t, p_c, m)$ (3).

where m is the shared backbone model used across all experts. The experts group output is defined as: $y_g = \{y_c \mid c \in y_r\}$ (4).

Orchestrator To finalize the process, we design an Orchestrator Agent A_o to summarize the outputs of the Figurative Explainer Agent and all activated Specialized Agents into a unified final report for tasks T1 ~T8. This agent consolidates multi-perspective analyses into an interpretable, task-specific summary of the meme: $y_o = A_o(y_g, y_f, x_v, x_t, p_o, m)$ (5).

To streamline notation, we summarize the full procedure described in Equations 1 ~5 as a single function. We describe the technical details for our agentic framework in Algorithm 2. Specifically, we denote the zero-shot, multifaceted analysis of a meme $x = (x_v, x_t)$ performed by **AiO-MMAGents** with backbone model m as: $\{y_f, y_r, y_g, y_o\} = \text{MMAGents}(x, m)$ (6).

3.2 Human-AI Collaboration for Annotation

Human-AI Annotation has attracted increasing interest in recent work Zhang et al. [2025], Qian et al. [2025], Shen et al. [2025], where agentic generated responses is explored as a way to reduce reliance on costly manual labeling, and human verification can further improve annotation quality Han et al. [2025]. Our proposed Human-AI collaboration for annotation consists of two stages: 1) We introduce a Manager Agent A_{mgr} to consolidate two AiO-MMAGents teams’ outputs. The Manager Agent is powered by GPT-4o-mini as backbone model and guided by CoT prompting p_{mgr} . This agent integrates both teams’ outputs with background knowledge Tang et al. [2023] to produce high-quality multi-task annotations. Here, background knowledge refers to the subset of labels already available in source datasets; these labels are used to maintain consistency on covered tasks, but they do not

replace reasoning for missing tasks or newly introduced labels. We describe the details of automatic annotation with the Manager Agent in Algorithm 1 (Appendix B.3). 2) Human Verification for quality-control: domain experts review the synthesized outputs to attain high-quality labels, resolve inter-agent conflicts, discard low-confidence or unreliable predictions, and introduce new labels when existing annotations are insufficient. This human-AI collaborative design combines the scalability and consistency of agent-based automation with human judgment and domain awareness, ensuring high-quality and extensible annotations. The Human-AI collaboration produces a high-quality dataset, **Multifacet-MM**, consisting of four subsets: D_f , D_r , D_g , and D_o , which correspond to the Figurative Agent, Router, Experts Group, and Orchestrator Agent, respectively.

3.3 Agentic Distillation

Warm Start with Reasoning SFT To enable agentic optimization, we warm-start the policy model via reasoning fine-tuning. Specifically, as shown in Figure 1, we construct reasoning chains D_{rc} from our proposed Multifacet-MM dataset. In each reasoning chain rc , we first performs visual figurative analysis, then executes a router planning step to select expert agents for subsequent analysis. The selected experts examine the input meme from different social identity perspectives. Finally, a `<final>` step aggregates the preceding reasoning chain and outputs a structured prediction as the final answer. We fine-tune a student model $\pi_{\theta_{\text{SFT}}}$ on the constructed reasoning chains using LoRA Hu et al. [2021]. We denote the reasoning-chain dataset as $\mathcal{D}_{rc} = \{(x_i, rc_i)\}_{i=1}^N$. and define θ_{SFT} by

$$\theta_{\text{SFT}} = \arg \min_{\theta} \mathbb{E}_{(x, rc) \sim \mathcal{D}_{rc}} [-\log \pi_{\theta}(rc | x)] \quad (7)$$

Agentic Optimization with Learnable Skills 1) Based on GRPO Shao et al. [2024], we design rubric-based reward functions with verifiable rules that explicitly check structural compliance (e.g., required `<think>` and `<final>` blocks, mandatory keys) and constrain predictions to predefined label sets. We denote the rubric-based reward as $R_r(o) = \sum_{i=1}^m w_i G_{\text{rule}}(o, r_i)$, where G_{rule} is the grader, w_i is the weight of rule r_i , and o is the reasoning chain generated by the SFT student model, $o = \pi_{\theta_{\text{SFT}}}(x)$.

2) We build a learnable skills library $\mathcal{S}_{\psi}$ by fine-tuning MemeCLIP Shah et al. [2024] for the classification tasks of Intention (T5), Intensity (T6), and Attack Type (T7), as our preliminary results show that MemeCLIP is both efficient and high-performing after pretraining. These skill agents are triggered by the keyword `</final>`, and their MemeCLIP-based predictions are inserted into (and replace the corresponding fields in) o . We denote the skill-augmented process as $o' = \mathcal{S}_{\psi}(o)$, if `</final>` $\in o$. This internal MemeCLIP-based skills library is distinct from the standalone MemeCLIP baseline in Table 3, which is evaluated directly on its supported tasks without reasoning-chain generation or agentic optimization.

3) We measure semantic agreement with gold annotations using exact matching and set-based similarity (e.g., Jaccard overlap), producing a stable and interpretable reward signal R_g aligned with the target reasoning format. In parallel, we generate a supervision signal R_s to update the learnable MemeCLIP-Skills library toward the golden labels for tasks T5–T7 (Intention, Intensity, and Attack Type). Concretely, MemeCLIP-Skills $\mathcal{S}_{\psi}$ is trained in a classification regime:

$$\psi \leftarrow \psi - \eta_s \nabla_{\psi} \mathcal{L}_{\text{cls}}(o', y_{T5:T7}), \quad (8)$$

where parameters $y_{T5:T7}$ are the golden labels for tasks T5–T7. The student policy is further optimized with a GRPO-style objective under the combined reward $R \triangleq R_r + \lambda R_g$. Specifically, we maximize a clipped policy-gradient objective:

$$\theta \leftarrow \arg \max_{\theta} \mathbb{E} \left[\sum_t \min \left(r_t(\theta) \hat{A}_t, \tilde{r}_t(\theta) \hat{A}_t \right) \right] \quad (9)$$

where s_t denotes the partial reasoning-chain context at step t , $r_t(\theta) \triangleq \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$, and a_t is the next generated token (action) under the policy. $\hat{A}_t$ is the advantage estimate computed from reward R , $\tilde{r}_t(\theta) \triangleq \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)$.

4 Experimental Setup

Datasets We evaluate our approach on two public datasets, **HatReD** Hee et al. [2023] and **MET-Meme** Xu et al. [2022]. The former focuses on harmful meme analysis, annotated with social category, social target, attack type, and reasoning. Additionally, we merge label information (social category and attack type) from WOA-5 Mathias et al. [2021], which shares overlapping memes with HatReD, originally sourced from the Hateful Memes Challenge Kiela et al. [2020]. The latter targets general-topic, multifaceted meme understanding with a focus on metaphor, sentiment, intention, and intensity. We randomly sample 3,300 memes from the two datasets (1,650 each), enabling a comprehensive evaluation of our work across both harmful and general-topic memes.

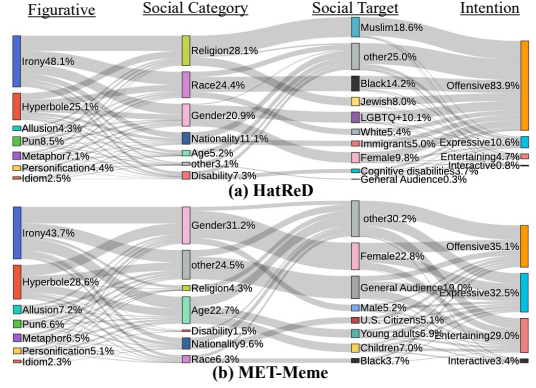


Figure 2: Label distributions of **Multifacet-MM** across Figurative (T1), Social Category (T2), Social Target (T3), and Intention (T5).

Notably, each meme in our dataset is annotated across 8 distinct tasks, whereas prior studies typically employed only 2 or 4. This results in a total of 26,400 annotation instances, making our dataset the second largest by total task-sample pairs.

Baselines 1) We evaluate the zero-shot performance of AiO-MM Agents using five different backbone models: GPT-4o-mini, Qwen2.5-VL-72B Bai et al. [2023], Qwen2.5-VL-7B Bai et al. [2023], LLaVA1.6-vicuna-13B Liu et al. [2023], and LLaMA3.2-11B-Vision Grattafiori et al. [2024]. This selection allows us to compare both closed- and open-source models, as well as large- and small-scale MLLMs. 2) To evaluate AiO-MM against strong supervised alternatives, we include three multi-task SFT baselines with different architectural forms: MemeCLIP, LLaMA3.2-11B-workflow, and Qwen3-VL-8B Reasoning SFT.

3) We use GPT-4o-mini as the backbone for the Manager Agent in the Human-AI Annotation stage (Section 3.2). 4) AiO-MM (Ours): We use the same Qwen3-VL-8B student backbone and apply the full agentic distillation pipeline, including reasoning SFT warm start, GRPO-style optimization, and the learnable skills library.

Table 3: Performance comparison among zero-shot AiO-MM Agents, the SFT LLaMA3.2-11B-workflow, 3-shot CoT and our proposed AiO-MM. Default metric is F1-score (%). [†] Social Target & Explanation uses BERTScore-F1 (%) to emphasize semantic similarity.

	Backbone Model	Method	Figurative		Router		Orchestrator					Avg
			Figurative	Social Category	Sentiment	Intention	Intensity	Attack Type	Social Target [†]	Explanation [†]		
HatReD	GPT-5.1	3-shot CoT	29.41 (0.66)	66.04 (1.02)	30.32 (1.02)	82.25 (0.92)	73.58 (1.09)	40.50 (1.01)	85.57 (0.12)	88.85 (0.04)	62.06	
	GPT-4o	3-shot CoT	35.96 (0.62)	56.12 (1.11)	22.48 (0.96)	74.46 (1.06)	68.91 (1.14)	34.01 (0.94)	87.62 (0.16)	89.28 (0.04)	58.60	
	GPT-4o-mini	3-shot CoT	36.23 (0.63)	43.49 (1.14)	42.82 (1.11)	78.78 (0.99)	69.33 (1.14)	34.03 (0.94)	87.80 (0.17)	89.34 (0.05)	60.23	
	Qwen2.5-VL-72B	3-shot CoT	39.62 (0.54)	52.30 (1.13)	50.43 (1.11)	68.67 (1.13)	33.45 (1.16)	29.73 (0.92)	86.76 (0.15)	88.09 (0.10)	56.13	
	Qwen3-VL-Max	3-shot CoT	55.33 (0.46)	68.05 (1.34)	32.15 (0.95)	79.85 (1.24)	72.15 (1.21)	25.50 (1.02)	85.78 (0.11)	87.65 (0.08)	63.32	
	GPT-4o-mini	MMAGents	45.19 (0.63)	56.04 (1.02)	71.46 (0.97)	74.17 (1.06)	73.09 (1.09)	40.85 (1.01)	92.32 (0.19)	90.58 (0.08)	67.96	
	Qwen2.5-VL-72B	MMAGents	78.46 (0.84)	60.48 (1.05)	68.97 (0.99)	69.98 (1.11)	67.15 (1.16)	43.36 (1.02)	92.07 (0.21)	88.82 (0.11)	71.16	
	Qwen2.5-VL-7B	MMAGents	32.76 (0.79)	41.51 (1.04)	32.00 (1.02)	56.03 (1.20)	22.55 (1.03)	24.52 (0.82)	88.67 (0.19)	88.15 (0.05)	48.27	
	LLaVA1.6-vicuna-13B	MMAGents	31.46 (0.72)	45.26 (0.97)	10.80 (0.70)	32.57 (1.14)	20.48 (0.99)	12.34 (0.66)	85.51 (0.16)	85.54 (0.09)	40.50	
	LLaMA3.2-11B-Vision	MMAGents	35.38 (0.61)	35.44 (1.01)	29.26 (1.02)	56.53 (1.21)	34.91 (1.17)	17.37 (0.73)	85.98 (0.16)	85.71 (0.09)	47.57	
MET-Meme	MemeCLIP	SFT	41.72 (2.74)	61.66 (2.70)	37.54 (2.72)	82.51 (1.76)	72.43 (2.85)	44.53 (2.18)	-	-	56.73	
	LLaMA3.2-11B-workflow	SFT	49.32 (0.80)	56.98 (0.94)	58.14 (0.98)	81.66 (0.92)	66.67 (1.16)	44.65 (0.88)	90.85 (0.20)	89.57 (0.05)	67.23	
	Qwen3-VL-8B	Reasoning SFT	56.05 (1.80)	66.84 (2.28)	41.81 (2.44)	72.67 (2.58)	57.67 (2.86)	25.67 (2.26)	89.55 (0.73)	87.69 (0.09)	62.24	
	AiO-MM (Ours)	Agentic Distillation	58.29 (2.58)	70.43 (2.85)	40.17 (2.44)	84.66 (1.91)	73.31 (2.39)	45.40 (2.65)	90.31 (1.05)	88.00 (0.31)	68.82	
	GPT-5.1	3-shot CoT	37.33 (0.79)	81.73 (0.92)	46.78 (1.03)	51.63 (1.21)	59.27 (1.06)	83.32 (0.89)	85.39 (0.10)	88.63 (0.03)	66.76	
	GPT-4o	3-shot CoT	38.39 (0.74)	80.54 (0.95)	38.16 (1.01)	46.72 (1.24)	58.00 (1.22)	84.90 (0.85)	86.76 (0.14)	89.14 (0.04)	65.33	
	GPT-4o-mini	3-shot CoT	33.33 (0.74)	80.22 (0.96)	38.13 (1.03)	52.49 (1.18)	51.45 (1.23)	85.45 (0.84)	87.01 (0.14)	89.41 (0.04)	64.69	
	Qwen2.5-VL-72B	3-shot CoT	41.81 (0.71)	80.16 (0.96)	43.34 (1.01)	48.64 (1.15)	32.91 (1.16)	76.61 (1.02)	85.92 (0.12)	88.23 (0.09)	62.20	
	Qwen3-VL-Max	3-shot CoT	59.23 (0.73)	82.42 (1.02)	36.67 (0.95)	50.60 (1.24)	43.57 (1.16)	59.27 (1.08)	82.04 (0.16)	87.87 (0.07)	62.71	
	GPT-4o-mini	MMAGents	53.92 (0.69)	86.02 (0.82)	55.53 (0.97)	54.14 (1.21)	69.70 (1.13)	89.35 (0.73)	96.61 (0.14)	91.48 (0.08)	74.59	
Qwen2.5-VL-72B	MMAGents	69.44 (0.86)	81.12 (0.93)	53.76 (0.96)	54.57 (1.19)	63.27 (1.19)	87.97 (0.78)	92.24 (0.18)	89.53 (0.09)	73.99		
Qwen2.5-VL-7B	MMAGents	33.74 (0.79)	69.06 (1.10)	16.76 (0.76)	40.96 (1.18)	37.94 (1.19)	74.92 (1.04)	87.51 (0.15)	88.12 (0.06)	56.13		
LLaVA1.6-vicuna-13B	MMAGents	32.24 (0.74)	16.69 (0.82)	9.59 (0.62)	21.87 (1.00)	29.15 (1.12)	46.98 (1.22)	84.68 (0.13)	85.68 (0.09)	40.86		
LLaMA3.2-11B-Vision	MMAGents	33.12 (0.71)	14.35 (0.80)	13.32 (0.74)	27.92 (1.09)	22.48 (1.03)	35.43 (1.17)	85.35 (0.13)	85.34 (0.09)	39.66		
MET-Meme	MemeCLIP	SFT	39.82 (2.74)	66.35 (2.20)	28.74 (2.36)	47.25 (2.80)	74.30 (2.50)	81.34 (1.28)	-	-	56.30	
	LLaMA3.2-11B-workflow	SFT	45.34 (0.80)	65.37 (1.11)	42.11 (0.92)	44.29 (1.19)	44.79 (1.22)	67.88 (1.11)	89.73 (0.17)	89.35 (0.05)	61.11	
	Qwen3-VL-8B	Reasoning SFT	57.59 (1.83)	79.36 (2.22)	26.50 (2.43)	39.74 (2.77)	52.24 (2.83)	70.41 (2.56)	88.96 (0.33)	87.78 (0.08)	62.82	
	AiO-MM (Ours)	Agentic Distillation	60.09 (2.91)	76.42 (2.51)	33.25 (2.08)	49.10 (2.70)	76.65 (2.30)	85.03 (1.90)	91.33 (0.97)	87.93 (0.20)	69.99	

Evaluation We use BERTScore-F1 Zhang et al. [2019] as the evaluation measure for generation tasks (T3 & T8), and standard F1-score for the remaining classification tasks. We provide more implementation details in Appendix C.1.

5 Results

In this study, we aim to answer the following 3 research questions:

RQ1: How effectively can AiO-MM Agents perform Multifaceted CMU? We present the zero-shot performance of the proposed agent framework across 8 tasks with 5 backbone models in Table 3. 1) We observe that AiO-MM Agents consistently outperform few-shot CoT baselines by a substantial margin across all eight tasks. For example, using Qwen2.5-VL-72B as the backbone model, AiO-MM Agents achieve an average score of 71.16%, compared to 56.13% obtained with few-shot CoT on the HatReD dataset. Statistical significance tests further confirm these improvements ($p < 0.05$). 2) Despite its best overall capabilities in few-shot CoT group, GPT-5.1 shows comparatively weaker performance in visual figurative reasoning than Qwen2.5-VL-72B and GPT-4o. Figurative language recognition remains a challenging task, consistently ranking among the top three hardest classification tasks across both datasets.

RQ2: To what extent can the agentic distilled student model AiO-MM replicate the performance of the full agentic framework’s on each subtask? As shown in Table 3, the supervised baselines in Table 3 all serve as multi-task SFT baselines, but from different modeling perspectives: MemeCLIP tests a meme-specific classifier on its supported tasks, LLaMA3.2-11B-workflow tests a task-wise SFT workflow, and Qwen3-VL-8B Reasoning SFT isolates the warm-start student before agentic optimization. Under this comparison, AiO-MM significantly outperforms MM Agents with small-scale MLLMs as backbones with $p < 0.001$.

While there remains a performance gap between zero-shot AiO-MM Agents with large-scale MLLMs like GPT-4o-mini and Qwen2.5-VL-72B, this is expected given the substantial difference in parameter scale and training resources. Interestingly, AiO-MM performance closely matches few-shot CoT with advanced MLLMs such as GPT-4o, GPT-4o-mini, and Qwen2.5-VL-72B on HatReD. This indicates that general meme understanding is more challenging. This result shows that Agentic Distillation can effectively infuse the knowledge of AiO-MM Agents into smaller student models with limited resources.

RQ3: How is figurative language used by meme creators in the context of multifaceted memes?

To answer this question, we analyze the distribution of meme labels in **Multifacet-MM** across 4 key tasks: Figurative Language Liu et al. [2022], Social Category and Social Target Hee et al. [2023], and Intention Xu et al. [2022].

These tasks collectively offer a unified view on meme understanding by integrating insights from prior multi-task work. The results are presented in Figure 2. We find that irony and hyperbole are the two dominant figurative expressions, appearing in over 72% of memes in HatReD and MET-Meme. Moreover, metaphor and personification occur in 11.5% of memes in HatReD and 11.6% in MET-Meme. This finding supports our hypothesis that a broader understanding of diverse figurative language is important for Multifaceted CMU. In contrast, prior studies have primarily focused on

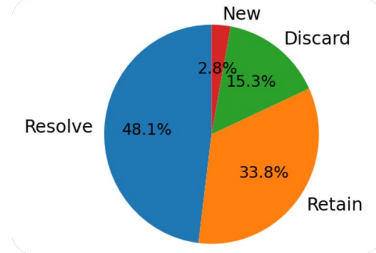
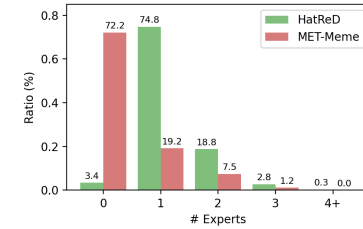
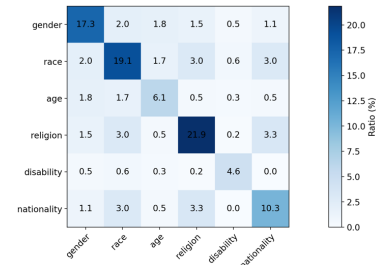


Figure 3: Human verification outcomes for the 26,400 task-sample pairs in Multifacet-MM.



(a) Distribution of activated experts.



(b) Co-occurrence of selected experts.

Figure 4: Expert routing behavior in MM Agents.

metaphor or considered only a small subset of figurative expressions Hwang and Shwartz [2023], Xu et al. [2024], Lin et al. [2024a], Chauhan et al. [2020]. Our study addresses this gap by exploring a richer taxonomy of 8 figurative expressions across both harmful and general memes. For further analysis, refer to Appendix B.1.

5.1 Analysis

Human Verification for Data Quality Control

Synthetic datasets generated by agentic frameworks may contain low-quality or conflicting annotations that are misaligned with human preferences. To address this issue, we propose a *Human-AI annotation* paradigm that incorporates human supervision after the Manager Agent aggregates final outputs from two MemeAgent teams. This design enables effective quality control by combining automated reasoning with human judgment.

As illustrated in Figure 3, among 26,400 task-sample pairs produced by the Manager Agent, 33.8% of the labels are identified as high-quality and directly retained. In addition, 48.1% of the synthetic labels require domain experts to resolve annotation conflicts, 15.3% are discarded due to low quality, and 2.8% correspond to newly introduced labels.

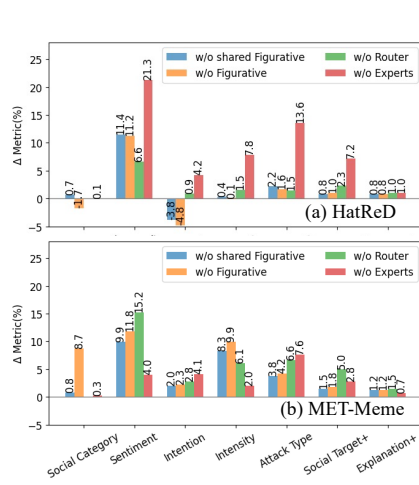


Figure 5: Ablation study with Qwen2.5-VL-72B. Each bar shows the performance drop.

Moreover, Figure 4b reveals frequent co-occurrence between multiple social-identity experts, indicating that many memes simultaneously target overlapping identity dimensions. Overall, this supports the necessity of our mixture-of-agents design and validates the motivation behind MMAGents as a flexible framework for multifaceted meme understanding.

Evaluation for Error Propagation. We conduct a *forced-setting* experiment in which downstream tasks are provided with ground-truth outputs from the Figurative Agent (i.e., assuming 100% figurative understanding accuracy), using Qwen2.5-VL-72B as the backbone model. We define the *Relative Impact Ratio* as: $\text{Relative Impact} = \frac{\Delta_{\text{sub-task}}}{\Delta_{\text{Figurative}}}$, where Δ denotes the change in task performance shown in Table 3 under the *forced-setting*.

The Relative Impact Ratio quantifies whether errors originating from figurative understanding are amplified or attenuated along the agent collaboration pipeline. As shown in Figure 6, Relative Impact ratios below 1 across tasks and domains indicate that error propagation is generally limited under the proposed MMAGents framework. The largest relative impact ratio is 82.2% for the Attack Type Classification task on the HatReD dataset, suggesting that accurate figurative understanding can contribute to identifying attack types. This observation aligns with human intuition, as recognizing underlying figurative cues is often critical for correctly interpreting the intent and nature of harmful memes.

To further assess the robustness of the Router, we apply the same forced-setting methodology, this time assuming 100% routing accuracy (i.e., all relevant experts are activated). As shown in Figure 7, the average Relative Impact across tasks and domains is 28.73%, confirming that routing errors also have limited downstream effects overall. The most sensitive task is Sentiment on MET-Meme (86.1%), indicating that routing to the correct social-category experts is particularly important for sentiment prediction in general memes. In contrast, Social Target and Explanation tasks show consistently low sensitivity (below 14%) across both domains, as these outputs are largely determined by the Figurative Agent and Orchestrator regardless of which expert paths are activated.

Ablation Study.

We conduct an ablation study to assess the contribution of each component in AiO-MMAGents, using Qwen2.5-VL-72B as the backbone.

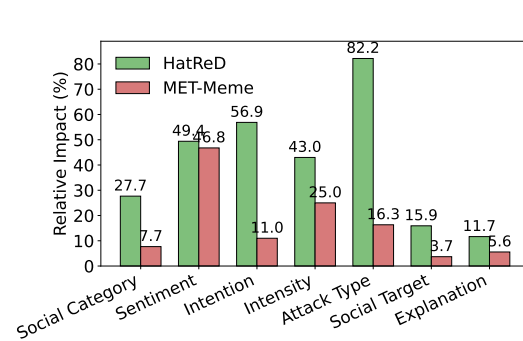


Figure 6: Relative Impact (%) of error propagation from the Figurative Agent on downstream tasks across HatReD and MET-Meme. The corresponding Router error propagation results are shown in Figure 7 (Appendix C.4).

its shared signals improves performance on Social Category and Intention in HatReD, but this effect does not generalize to MET-Meme, where these components instead improve Social Category, Sentiment, and Intensity. This difference may stem from HatReD memes relying more on direct expressions of intent toward social groups. Overall, integrating all components in AiO-MMAGents supports robust generalization across harmful and general meme datasets, leading to strong performance on Multifaceted CMU.

We evaluate the following variants: (a) *w/o* shared Figurative, which removes the shared output from the Figurative Agent; (b) *w/o* Figurative, which removes the Figurative Agent entirely; (c) *w/o* Router, which forwards all memes to all Expert agents; and (d) *w/o* Experts, which removes the Experts Group. Results are shown in Figure 5.

The impact of individual components varies across datasets. On HatReD, the Experts Group contributes broadly, yielding gains in Sentiment Classification (+21.3%) and Attack Type Detection (+13.6%).

On MET-Meme, different components dominate different tasks: the Router contributes most to Sentiment (+15.2%), while the Figurative Agent provides the largest gain in Intensity (+9.9%). Interestingly, removing the Figurative Agent or

6 Conclusion & Future Work

In this work, we provided a comprehensive analysis of the multifaceted characteristics of memes, highlighting the important role of figurative language in improving multi-task meme understanding. We introduced AiO-MMAGents, an All-in-One multi-agent framework for Multifaceted CMU. We also curated Multifacet-MM, a high-quality, automatically annotated dataset with agents collaboration. We distilled a high performance workflow with small-scale student models, AiO-MM. Experiments demonstrated the effectiveness and generalizability of our work, and show that agentic distillation can effectively empower student models to replicate the reasoning patterns of AiO-MMAGents.

7 Limitations

While our proposed pipeline shows high performance and effectiveness in Multifaceted CMU, several limitations remain. First, although we cover 8 distinct tasks of meme understanding, there are other tasks yet to be explored. For example, integrating persuasive techniques Hasanain et al. [2023], Dimitrov et al. [2024] detection could further enrich the capability of our framework. Additionally, the ambiguity of multimodal figurative expressions in memes increases annotation difficulty Ziegler et al. [2020]. Future work can explore strategies to address this issue.

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642 A Prompts Details

643 A.1 Prompt engineering strategies

644 The related studies that informed our prompt design are as follows: 1) Figurative agent: Xu et al.
645 [2022] inspired our prompt for visual metaphor analysis, and Liu et al. [2022] encouraged us to
646 further extend the scope to general figurative expressions. 2) Router: DeepSeek-AI et al. [2024]
647 inspired us to incorporate a router agent to avoid unnecessary computation. 3) Experts Group: Hee
648 et al. [2023] motivated us to create a group of agents representing different social categories for meme
649 analysis.

650 A.2 Figurative Agent

651 You are an expert at analysis and explanation of the figurative expressions used in Memes. **ROLE:**
652 Identify figurative expressions in the meme text/description (at most 2 types). Optional figurative
653 categories are: {Figurative Language Taxonomy}

654 A.3 Router

655 You are a Router responsible for selecting which aspects should be censored to better understand a
656 meme's sentiment, intention, intensity, attack type, and target group.

657 **Task:** Given a meme (text + image) and figurative language expert explanation for the meme,
658 determine which of the following aspects should be censored to enhance interpretability: Social
659 Category list Your decision should be based on: 1) Target Relevance. Does the meme explicitly
660 or implicitly reference any of these aspects? 2) Sentiment & Intensity Impact. Does censoring a
661 particular aspect help better analyze how strong the sentiment or attack is? 3) Contextual Necessity.
662 Is the aspect central to understanding the meme?

663 A.4 Expert Agents

664 You are the Activated Social Category Agent. **ROLE:** Classify how this meme addresses or affects
665 individuals/groups based on race or ethnicity. **EXAMPLE GROUPS** Example list of Social Target
666 options

667 **TASK REQUIREMENTS** 1) Determine sentiment, intention, intensity, attack type, and target group
668 if relevant. a) Sentiment: Identify the emotion that the creator intends to evoke in the audience,
669 options [happiness, love, anger, sorrow, fear, hate, surprise, neutral] b) Intention: Determine the
670 purpose or motivation behind creating the meme, options [interactive, expressive, purely entertaining,
671 offensive, other] c) Intensity: Assess the strength or degree of the sentiment being conveyed. A
672 numerical score from 1 to 4, where 1 = very mild or weak, 4 = very strong or intense. d) Target Group:
673 Identify the intended audience demographic the meme is created for, options Social target options e)
674 Attack Type: If the meme contains harmful or offensive content, specify the type of attack being used,
675 options [dehumanizing, inferiority, inciting_violence, mocking, contempt, slurs, exclusion, none] 2)
676 If no race-based target, "None". 3) Keep reasoning concise and within 30 words. 4) The Attack Type
677 category applies exclusively to hateful or offensive memes; in all other cases, it should be marked as
678 "None."

679 A.5 Orchestrator Agent

680 You are the Aggregator/Orchestrator. **ROLE:** Combine outputs from the Metaphor Explanation
681 Agent and each Specialized Agent into a final report. Multifaceted CMU definition

682 **TASK REQUIREMENTS:** 1) Produce a single JSON report that includes: a) An aggregated
683 conclusion: b)possibly identify any dominant target group(s) c) overall sentiment, intention, intensity,
684 and attack type (if relevant) d)note that memes can serve multiple purposes or intentions (not always
685 hateful) 2) Keep the final reasoning concise. 3) The Attack Type category applies exclusively to
686 hateful or offensive memes; in all other cases, it should be marked as "None."

687 A.6 Manager Agent

688 You are a manager agent. Based on the task guidelines, background label information from existing
689 datasets, and results from sub-agents, generate the final output. Use the background label information
690 only as consistency guidance for tasks already covered by source datasets, while relying on reasoning
691 over the meme and sub-agent outputs to infer missing tasks and produce new labels when needed.
692 Task guideline: Agents prompt Input Meme captions: Caption. Background label information:
693 Background information. Result from sub-agents: outputs from other AiO-MMAgents teams.

694 A.7 CoT

695 **Instructions:** Given a meme (image and caption), answer each of the following 8 questions with a
696 single, clear choice or response. Output your results as a JSON object.

697 CoT Questions (One per Task):

698 T1 (Figurative): Which types (up to two) of figurative language is most prominent in the meme?
699 Options: "metaphor", "irony", "hyperbole", "onomatopoeia", "personification", "allusion", "idiom",
700 "pun", "none"

701 T2 (Social Category): Which social category is primarily referenced or targeted by the meme?
702 Options: "gender", "race", "age", "religion", "disability", "nationality", "none"

703

704 T7 (Attack Type): If the meme is offensive or harmful, what are the primary attack types (up to
705 two)? Options: "dehumanizing", "inferiority", "inciting violence", "mocking", "contempt", "slurs",
706 "exclusion", "none"

707 T8 (Explanation): Briefly explain your choices above (1–2 sentences).

708 Note: For each task with a list of options, please ensure your answer is selected from that list.

709 B Methodology Details

710 B.1 Figurative Language Taxonomy

711 We consider the following figurative language types according to the knowledge from Wikipedia
712 page: https://en.wikipedia.org/wiki/Literal_and_figurative_language.

- 713 • Metaphor: A comparison between two unlike things. e.g., "Time is a thief."
- 714 • Onomatopoeia: Words that imitate natural sounds. e.g., "buzz," "crash," "sizzle," "bang."
- 715 • Personification: Giving human traits to non-human things or ideas. e.g., "The wind whis-
716 pered through the trees."
- 717 • Irony: Irony is when there's a contrast between expectation and reality — what is said or
718 expected is different from (or opposite to) what actually happens or is meant.
- 719 • Hyperbole: An exaggeration for emphasis or effect. e.g., "I'm so hungry I could eat a horse."
- 720 • Allusion: A reference to a well-known person, place, event, or work of art/literature. e.g.,
721 He met his Waterloo." (Alludes to Napoleon's defeat).
- 722 • Idiom: A common phrase whose meaning is different from its literal meaning. e.g., "Kick
723 the bucket" = to die; "Break the ice" = to start a conversation.
- 724 • Pun: A play on words, often using words that sound alike or have multiple meanings. e.g.,
725 "Time flies like an arrow; fruit flies like a banana."

726 B.2 Social Target Taxonomy

727 The Social Target taxonomy supports nuanced detection of social entity specific harms. We list a few
728 examples as the target Entities for each social category, and prompt the agent to either generate the
729 most proper fine-grained target group or choose the most suitable one from the pre-defined list of
730 each social category:

Algorithm 1 Manager Agent

Input: A meme dataset D , A Manager Agent A_{mgr} , AiO-MMAGents and its prompts set $\{p_f, p_r, p_c, p_o\}$ and 2 backbones m^a, m^b
Output: Multifacet-MM Dataset $\{D_f, D_r, D_g, D_o\}$
Initialize $\{D_f, D_r, D_g, D_o\} \leftarrow \{\emptyset, \emptyset, \emptyset, \emptyset\}$
for each meme $x \in D$ **do**
 // zero-shot AiO-MMAGents (Eq. 6)
 $\{y_f^a, y_r^a, y_g^a, y_o^a\} \leftarrow \text{MMAGents}(x, m^a)$
 $\{y_f^b, y_r^b, y_g^b, y_o^b\} \leftarrow \text{MMAGents}(x, m^b)$
 Get background information of x as k
 // Annotate Figurative Agent response
 $y_f \leftarrow A_{mgr}(y_f^a, y_f^b, x, k, p_f)$
 // Figurative response dataset
 Add (x, p_f, y_f) to D_f
 // Annotate Router response
 $y_r \leftarrow A_{mgr}(y_f, y_r^a, y_r^b, x, k, p_r)$
 Add (x, p_r, y_r) to D_r
 // Iterate over activated Experts
 for each category $c \in y_r$ **do**
 Fetch the mapped prompt p_c from $\mathcal{P}_C$
 // Annotate Expert response
 $y_c \leftarrow A_{mgr}(y_f, y_g^a, y_g^b, x, k, p_c)$
 Add (x, p_c, y_c) to D_g
 Add y_c to y_g
 end for
 // Annotate Orchestrator response
 $y_o \leftarrow A_{mgr}(y_f, y_g, y_o^a, y_o^b, x, k, p_o)$
 Add (x, p_o, y_o) to D_o
end for
return $\{D_f, D_r, D_g, D_o\}$

- 731 • **Gender:** Male, Female, LGBTQ+.
- 732 • **Race:** Black, White, Hispanic, Asian, Indigenous, etc.
- 733 • **Religion:** Christian, Muslim, Jewish, etc.
- 734 • **Disability:** Physical, sensory, cognitive, neurodivergent, etc.
- 735 • **Age:** Children, Teenagers, Young Adults, Middle-Aged Adults, Older Adults, etc.
- 736 • **Nationality:** Citizens, Immigrants, Refugees, Visitors, etc.

737 B.3 Manager Agent details

738 We present algorithm for Manager Agent in Alg 1 and MMAGents in Alg 2. Existing datasets cover
739 only subsets of the label space required for Multifaceted CMU (Table 1), we treat available labels as
740 background knowledge k to guide the annotation of missing tasks. Tang et al. [2023] Concretely, k is
741 not used as a supervision target for the entire output; instead, it provides consistency constraints on
742 already-covered tasks while leaving the Manager Agent free to reason about uncovered tasks and to
743 introduce new labels when the source annotations are incomplete.

744 C Annotation Protocol

745 We recruited 7 participants through institutional mailing lists and LinkedIn. Four self-identified as
746 experts in meme analysis and three as intermediate practitioners. They were split into two groups:
747 3 annotators (P1–P3) participated in Human–AI verification, and 4 annotators (P4–P7) conducted
748 independent evaluation. All annotators completed a qualification test of 20 meme questions and
749 were required to achieve at least 90% accuracy before participating in the study. Table 4 summarizes
750 annotator roles and time cost across the Human–AI verification and independent evaluation stages.

Algorithm 2 AiO-MMAGents

Input: A meme x consisting of image x_v and caption x_t ; a multimodal LLM m ; an agent set $\{A_f, A_r, A_g, A_o\}$ with corresponding prompts $\{p_f, p_r, p_c, p_o\}$
Output: A set of agent outputs Y
Initialize $Y \leftarrow \emptyset, y_g \leftarrow \emptyset$
// Figurative Agent
 $y_f \leftarrow A_f(x_v, x_t, p_f, m)$
 $y_r \leftarrow A_r(y_f, x_v, x_t, p_r, m)$ *// Router*
// Iterate over Experts activated by Router
for each category $c \in y_r$ **do**
 Fetch the mapped prompt p_c from $\mathcal{P}_C$
 // Category-specific Expert
 $y_c \leftarrow A_g(y_f, x_v, x_t, p_c, m)$
 Add y_c to y_g
end for
// Orchestrator Agent
 $y_o \leftarrow A_o(y_g, y_f, x_v, x_t, p_o, m)$
 $Y \leftarrow \{y_f, y_r, y_g, y_o\}$
return Y

Table 4: Annotator roles and time cost in the Human–AI verification and independent evaluation stages.

PID	Meme expertise	Time cost	Position	Verification	Evaluation
P1	Expert	30h	PhD	Y	N
P2	Intermediate	30h	Undergrad	Y	N
P3	Expert	4.5h	PhD	Y	N
P4	Expert	2h	PhD	N	Y
P5	Intermediate	2h	PhD	N	Y
P6	Intermediate	2h	PhD	N	Y
P7	Expert	2h	PhD	N	Y

During verification, annotators reviewed Manager-Agent outputs together with the meme and the source-agent traces. They retained high-confidence outputs, resolved conflicts between the two agent teams, discarded low-quality predictions, and introduced new labels when existing task definitions were insufficient. This process yielded the distribution summarized in Figure 3.

For independent evaluation, we randomly sampled 15% of the dataset, corresponding to 3,960 task–label pairs, and asked four qualified annotators to judge each pair independently with a binary agreement/disagreement decision. The satisfaction rate is the proportion of labels marked as agreement. We report Cohen’s kappa to quantify inter-annotator agreement over these judgments. Table 5 reports the per-domain results.

C.1 Implementation Details

We use the official APIs to access the large-scale MLLMs, GPT-4o-mini and Qwen2.5-VL-72B. The 3 small-scale MLLMs are hosted locally, with model parameters obtained from Hugging Face. We set the maximum output token limit to 400 and the temperature to 0.85 for all MLLMs. For Agentic Distillation, we fine-tune the student models using the Unsloth Daniel Han and team [2023] package with 16-bit LoRA and set rank size to 16. We adopt a 3-fold strategy and set epochs to 3. Other hyperparameters set with default values. All experiments are conducted on a Linux server with 3 NVIDIA RTX A6000 GPUs.

Table 5: Human evaluation results on the independently reviewed subset of Multifacet-MM.

Domain	Satisfaction rate (%)	Cohen’s κ
HatReD	87.24	0.69
MET-Meme	92.50	0.75
Avg	89.87	0.72

Table 6: Efficiency comparison across prompting, multi-agent, and distilled systems. API-based models do not require local deployment GPUs or training GPU hours.

Model	Method	Avg overall (%)	API cost (\$) / sample	Inference time (s) / sample	Deploy #GPU (A6000)	Training GPU hour
GPT-5.1	3-shot CoT	64.41	0.0079	6.09	-	-
GPT-4o-mini	3-shot CoT	62.46	0.0077	3.43	-	-
Qwen2.5-VL-72B	3-shot CoT	59.17	0.0091	6.82	-	-
GPT-4o-mini	MMAgents	71.28	0.0133	15.77	-	-
Qwen2.5-VL-72B	MMAgents	72.58	0.0155	17.08	-	-
LLaMA3.2-11B-Vision	MMAgents	43.62	-	130.18	1	-
LLaMA3.2-11B-workflow	SFT	64.17	-	117.30	3	4
AiO-MM (Ours)	Agentic Distillation	69.41	-	24.25	1	6

C.2 Efficiency Details

The full multi-agent pipeline improves quality at the cost of additional inference calls, but the overhead remains moderate relative to the observed gains. For example, with GPT-4o-mini as the backbone, MMAgents improves the overall average score from 62.46% to 71.28% while increasing the API cost from \$0.0077 to \$0.0133 per sample and latency from 3.43s to 15.77s. Our distilled AiO-MM retains a 69.41% average score while reducing local inference time to 24.25s on a single RTX A6000, substantially faster than the 117.3s required by the LLaMA3.2-11B workflow baseline. Full efficiency statistics are summarized in Table 6. The full MMAgents pipeline incurs additional inference cost because each meme is processed by a Figurative Agent, a Router, a subset of Experts, and an Orchestrator. In contrast, AiO-MM shifts this cost to a one-time distillation phase and substantially reduces deployment overhead at inference time.

C.3 Strategies to address the ambiguity of figurative expressions

The focus of this work is the agentic framework and agentic distillation for multifaceted meme understanding, not on an in-depth study of ambiguity inherent in figurative language. While there are already many works Chakrabarty et al. [2022], Kim et al. [2024] addressing this topic, we acknowledge that it remains an open problem and have highlighted it as an important limitation and a direction for future work.

C.4 Additional Figures

D Case Study

Figure 8 and Figure 9 present two representative examples illustrating how the agentic framework handles contrasting meme types. In Figure 8 (HatReD), the Figurative Agent identifies hyperbole—the exaggerated violent imagery combined with the pun “put the panic back in Hispanic”—as the core figurative device. Guided by this, the Router activates Race and Nationality experts, and the Orchestrator integrates these cues into an interpretable explanation of the meme’s hateful message. In Figure 9 (MET-Meme, Appendix), the Figurative Agent identifies pun and personification—combining “Hurricane Harvey” with a celebrity image to create a lighthearted joke. Since no hateful or identity-targeting cues are detected, the Router remains inactive, conserving computation while the Orchestrator correctly concludes the meme’s entertaining and benign intent.

Ethical Considerations

This work studies computational meme understanding using publicly available meme datasets and online content released for research purposes. The proposed framework focuses on analyzing

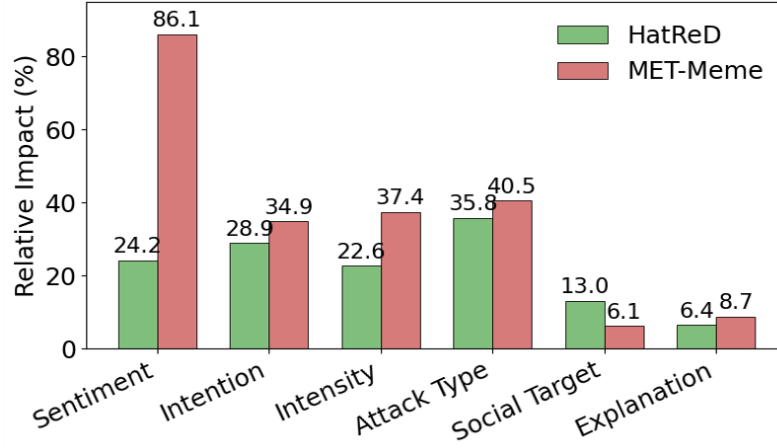


Figure 7: Router error propagation: Relative Impact (%) on downstream tasks across HatReD and MET-Meme. See Figure 6 for the corresponding Figurative Agent error propagation.

time to put the "panic"

Figurative Agent: Pun, Hyperbole

Router: [Race, Nationality]

Expert: *Race*

Social Target	Hispanic / Latinx
Sentiment	anger
Intention	offensive
Intensity	4
Attack Type	inciting violence

Expert: *Nationality*

Social Target	Hispanic / Latinx
Sentiment	anger
Intention	offensive
Intensity	4
Attack Type	Mocking, inciting violence

Orchestrator

Social Target	Hispanic / Latinx
Sentiment	anger
Intention	offensive
Intensity	4
Attack Type	['mocking', 'inciting_violence']
Explanation	The meme employs a pun to mock Hispanic identity while inciting violence through gun imagery. The negative stereotypes evoke strong anger, highlighting the offensive nature of the content.

Figure 8: Case study from HatReD: “time to put the ‘panic’ back in hispanic”.

799 multimodal signals (images and text) to support tasks related to figurative language, emotion, and
800 social harm, and does not involve interaction with human subjects or experiments on animals.

801 Human involvement in this work is limited to a low-risk evaluation and validation setting. Specifically,
802 qualified annotators assess the quality of automatically generated labels and agent outputs using
803 binary judgments. Annotators are not asked to produce original labels, disclose personal information,
804 or engage with private or sensitive data. All annotators participated voluntarily and evaluated publicly
805 available meme content.

806 Because memes may contain offensive, hateful, or sensitive material, we acknowledge the potential
807 psychological discomfort associated with exposure to such content. Annotators were informed of
808 this risk in advance and were allowed to discontinue participation at any time. We do not collect or
809 release personally identifiable information about individuals depicted in the memes, and we restrict
810 dataset usage to research purposes.

811 Our dataset construction relies on a Human–AI collaborative annotation process, combining agent-
812 generated outputs with human verification. To mitigate risks associated with incorrect, biased, or
813 misleading model outputs, we employ multi-agent cross-checking, background knowledge constraints,

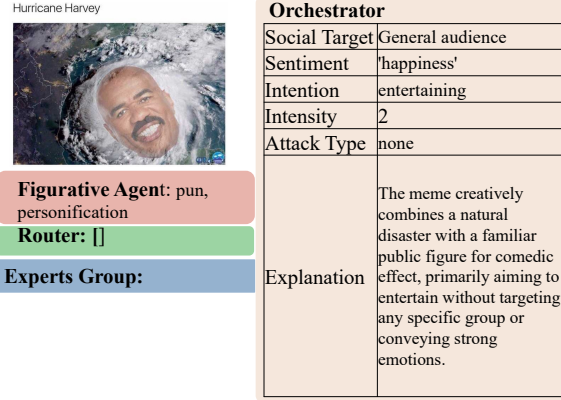


Figure 9: Case study from MET-Meme: “Hurricane Harvey”. The Figurative Agent identifies pun and personification. The Router remains inactive (no hateful cues), and the Orchestrator concludes the meme is entertaining and benign. See Figure 8 for the HatReD case study.

814 and human evaluation to improve label quality and transparency. We emphasize that the annotations
815 reflect interpretations of meme content rather than ground-truth intent, particularly for figurative and
816 ambiguous cases.

817 We also recognize that multi-agent LLM-based pipelines incur nontrivial computational cost. To
818 promote responsible use, we introduce an agentic distillation workflow that transfers reasoning
819 capabilities into smaller, more efficient student models, reducing inference-time resource consumption
820 and supporting more sustainable deployment.

821 Finally, the methods and datasets introduced in this work are intended for research and benchmarking
822 purposes. While our framework can assist in analyzing harmful content, it is not designed as a
823 standalone moderation or enforcement tool. We declare no conflicts of interest.

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